

ORCAST — Gate pipeline: detections to displayed confidence

Acoustic detections (OrcaHello)



L0 QC quarantine
(unreviewed stream isolated)



PSTH kernel fit
 $\log \lambda(x, t) = b_0 + \sum_k \beta_k \phi_k(t) + \log E$ (E1)



Phase-shuffled null test
 $p_{\text{null}} = P(T(\text{shuffled}) \geq T(\text{obs}))$ (E2)

↓ pass: $p < \alpha$

Time-rescaling GoF KS test
 $\tau_i = \Lambda(t_i) - \Lambda(t_{i-1}) \sim \text{Exp}(1)$ (E3)

↓ pass: KS ok

Held-out deviance skill
 $S_D = 1 - D_{\text{model}}/D_{\text{null}}$ (E5)

↓ pass: $S_D > 0$

PIT calibration KS test
 $u_i = F_{\text{model}}(y_i) \sim \text{Uniform}(0, 1)$ (E6)

↓ pass: calibrated

Effective confidence
 $c_{\text{eff}} = c_{\text{raw}} \cdot \prod_k g_k$ (E4)



Human promotion
(signed decision
record — DynamoDB)

--- fail ---

IC: no signal

Kernel contributes nothing; confidence unchanged

--- fail ---

IC: GoF failed

Model does not fit temporal spike train

--- fail ---

IC: skill negative

Model worse than null on held-out data

--- fail ---

IC: miscalibrated

$u_i \not\sim \text{Uniform}(0, 1)$; PIT fails



Gate PASS: all ICs clear
 $c_{\text{eff}} = c_{\text{raw}}$
(no gate reduced confidence)

Failing a gate surfaces an integrity condition alongside the metric. It does *not* suppress the forecast. The human promotion step is the final gate before confidence is raised. E1–E6: equation labels from the whitepaper.